

PrintConcrete: Isogeometric FEM simulator for concrete 3D printing

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The PrintConcrete IFEM simulator

- ▶ Quasi-static nonlinear FE simulator of the elasticity PDE

$$\nabla \boldsymbol{\sigma} + \mathbf{f} = \mathbf{0} \quad \text{where} \quad \boldsymbol{\sigma} = \mathbf{C}(t) : \boldsymbol{\varepsilon}(\mathbf{u})$$

where $\mathbf{C}$ represents a time-dependent material tensor, and $\mathbf{f}$ consists of the gravity load.

- ▶ The printing process is simulated by an element-birth feature in the solver, i.e., the stiffness and gravity loads for an element is set to zero until the birth time of the element.
- ▶ Material properties (Young's modulus E and yield stress limit σ_y) depend on the element age (time since element birth).
- ▶ The nonlinear PDE is linearized based on an Updated Lagrange formulation, and solved using Newton-Raphson iterations.
- ▶ Isogeometric analysis: Quadratic or cubic splines are used as basis functions in the FE implementation.
- ▶ The simulator is built on the IFEM-Elasticity package
<https://github.com/OPM/IFEM-Elasticity>.